

Grade 7 Accelerated
Unit 11
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

1. Austin works for a law firm and bills customers by the hour, below is a table of various customers he has worked with. How much money (in dollars) does Austin make per hour?

Time (hours)	Money earned (\$)
0.33	49.5
1.75	262.50
6.5	975
9	1,350

$$\frac{49.5}{0.33} = \$150/\text{hr}$$

2. Using the table above, if Austin works on average 37 hours per week, what is his weekly income?

$$150(37) = \$5,550 \text{ per week}$$

3. Does the following table show a proportional relationship between the weight of blue crab and price?

$$\frac{23.8}{4.25} = 5.6$$

$$\frac{11.03}{1.81} = 5.9$$

$$\frac{35.84}{6.4} = 5.5$$

Weight (lb.)	Price (\$)
4.25	23.80
1.87	11.03
6.4	35.84

No

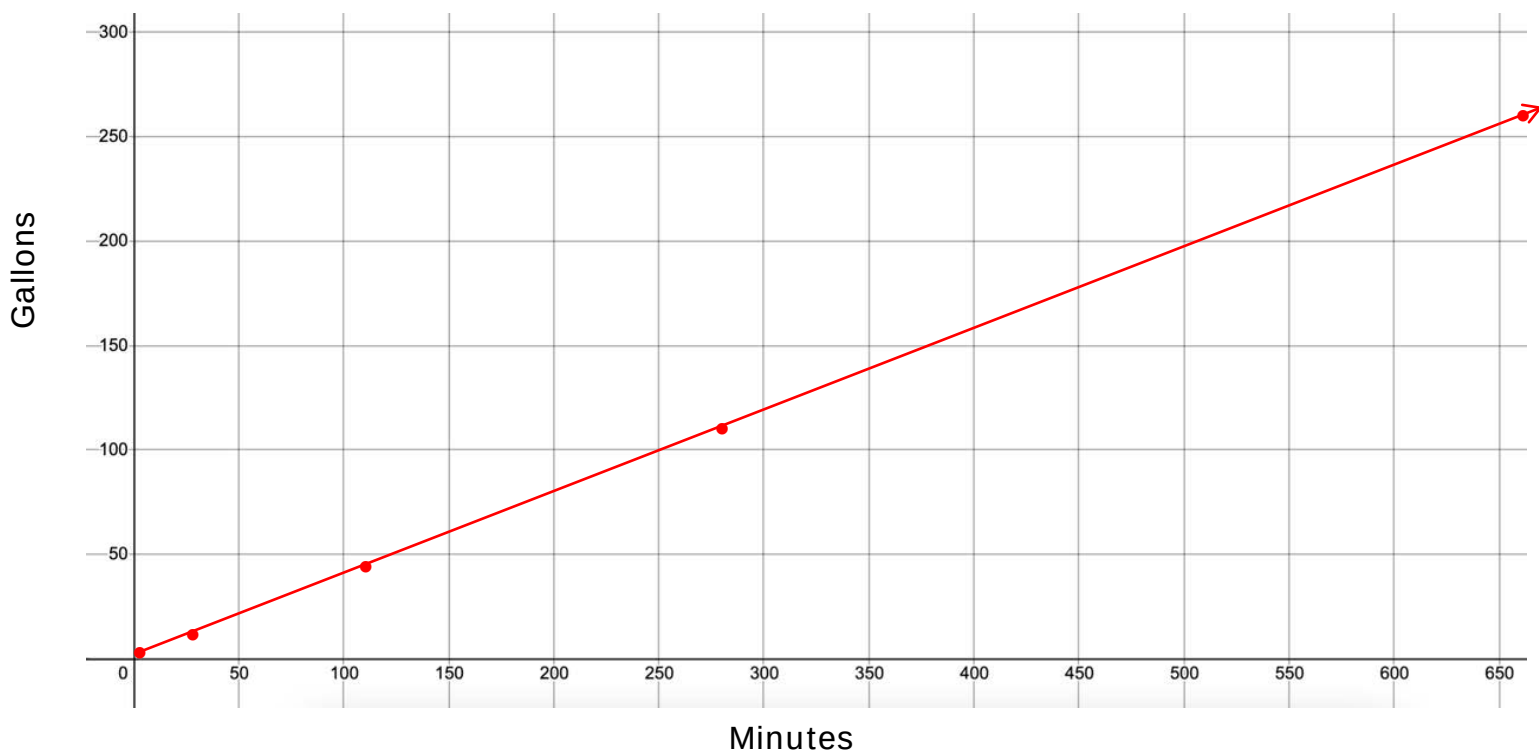
4. A hot tub is drained at a rate of 12 gallons every 30 minutes. If a standard hot tub holds 300 gallons of water, how long will it take to drain the hot tub?

$$\frac{12}{30} = \frac{300}{x} \quad \frac{12x}{12} = \frac{9000}{12} \quad x = 750 \text{ mins}$$

5. Using the information in question 4, complete the table below.

Time (in minutes)	Gallons Drained
2.5	1
30	12
105	42
660	264
280	112

6. Graph the proportional relationship below.



7. Looking at the graph, what are key indicators that the relationship between x and y values are proportional?

The points form a straight line that goes through the origin.